Problem

Title

Count Numbers with Exactly 3 Divisors

Topics

Divisor Counting, Prime Numbers, Nested Loops, Mathematical Insight, Conditional Logic

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a number theory problem-solving session, students are given a deeper mathematical observation.

The trainer asks:

"How many numbers from 1 to N have exactly **three divisors**?"

Students initially try brute-force:

1. For every number from 1 to N:
    - Count its divisors using a loop.
    - If divisor count equals 3 → increment counter.

But later, the trainer reveals an important mathematical insight:

A number has **exactly 3 divisors if and only if it is a perfect square of a prime number.**

Why?

If a number = p^2 (where p is prime),
Its divisors are:
1, p, p^2 → Exactly 3 divisors.

Examples:

- 4 → 1, 2, 4 ✓ (3 divisors)
- 9 → 1, 3, 9 ✓
- 25 → 1, 5, 25 ✓

- 16 → 1, 2, 4, 8, 16 ✖ (5 divisors)

Now, students are given an integer **N**.

They must determine how many numbers between **1 and N (inclusive)** have exactly three divisors.

Students may:

- Use brute-force divisor counting (nested loop).
- OR apply mathematical optimization (count prime numbers up to √N).
- Avoid built-in divisor functions.
- Implement proper prime checking using loops.

This problem strengthens:

- Divisor counting logic
- Prime number reasoning
- Mathematical pattern recognition
- Loop optimization thinking
- Time complexity awareness

Input Format

A single integer:

```
N
```

Constraints:

```
1 ≤ N ≤ 10^7
```

Output Format

- Print a single integer representing the count of numbers having exactly 3 divisors.

Examples

Example 1

Input:

```
10
```

Numbers with exactly 3 divisors:
4, 9

Output:

```
2
```

Example 2

Input:

```
25
```

Numbers:
4, 9, 25

Output:

```
3
```

Example 3

Input:

```
3
```

Output:

```
0
```